

spectively. You can also connect two more locks at D9 and D10 of IC2 along with relay driver transistors.

Once solenoid is activated, output remains latched and lock remains open (because it is a fail-

secure type lock). Reset the lock by switching S2 or S1 off and on again.

Construction and testing

An actual-size, single-side PCB pattern of the transmitter circuit is shown in Fig. 3 and its component layout in Fig. 4. An actual-size, single-side PCB pattern of the receiver circuit is shown in Fig. 5 and its component layout in Fig. 6. Both circuits require 12V regulated power supply. After assembling the circuit on the PCBs, enclose these in two separate plastic boxes. Before using the circuit, verify voltages are as per the test points given in the table.

Suppose you want to open solenoid lock1 connected across

PARTS LIST

Semiconductors:

- IC1 - HT12E encoder
- IC2 - HT12D decoder
- IC3 - 7805, 5V voltage regulator
- T1 - BC548 npn transistor
- T2, T3 - BD139 npn transistor
- LED1 - 5mm LED
- D1, D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-mega-ohm
- R2 - 47-kilo-ohm
- R3 - 1-kilo-ohm
- R4-R6 - 10-kilo-ohm

Capacitors:

- C1, C2, C4 - 100nF ceramic disk
- C3 - 100 μ F, 25V electrolytic

Miscellaneous:

- TX1 - 433MHz transmitter module
- RX1 - 433MHz receiver module
- CON1, CON2 - 2-pin terminal connector
- CON3, CON4 - 2-pin connector
- S1, S2 - On/off switch
- ANT.1, ANT.2 - 10cm-15cm-long single-strand wire antenna
- DIP1, DIP3 - 8-way DIP switch
- DIP2 - 4-way DIP switch
- TP0 - 12V solenoid lock1
- TP5 - 12V solenoid lock2
- TP6, TP7 - 12V regulated power supply/12V battery

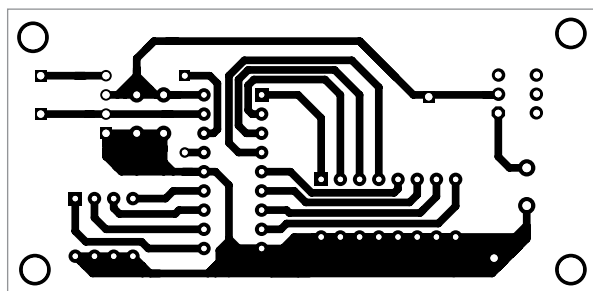


Fig. 3: Actual-size PCB pattern of the transmitter circuit

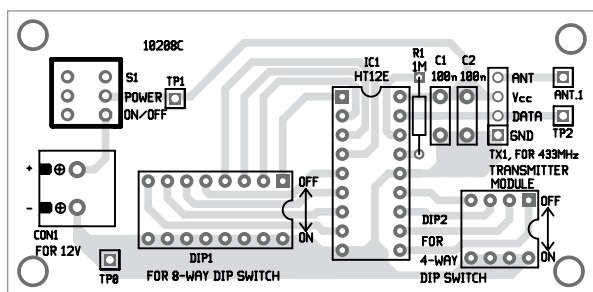


Fig. 4: Component layout of the PCB shown in Fig. 3

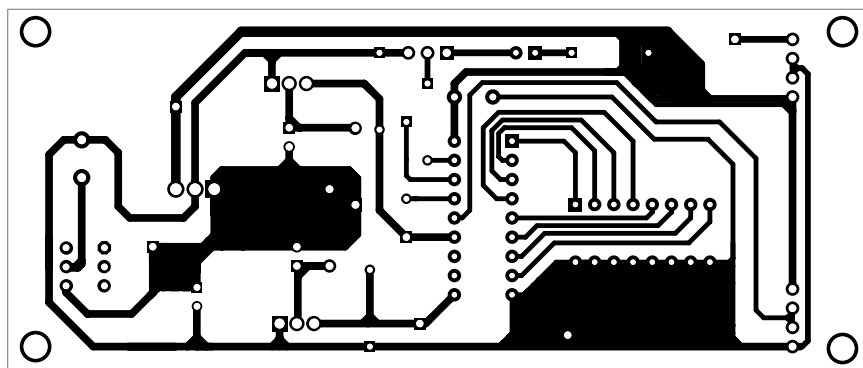


Fig. 5: Actual-size PCB pattern of the receiver circuit

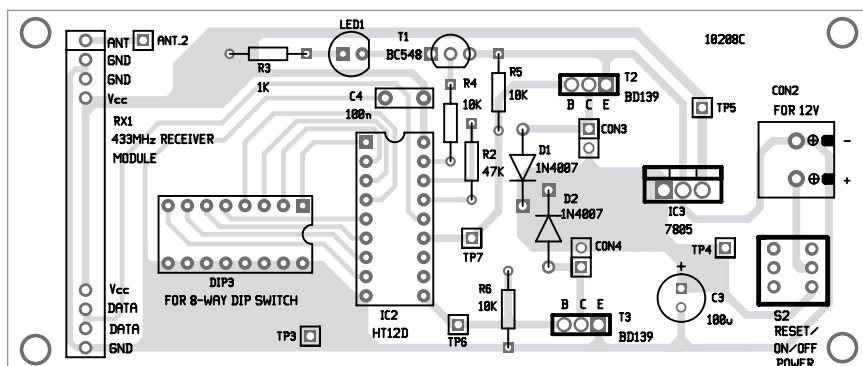


Fig. 6: Component layout of the PCB shown in Fig. 5

Test Points

Test point	Details
TP0, TP3	0V (GND)
TP1, TP4	12V provided switches S1 and S2 are on
TP2	Train of pulses
TP5	5V
TP6, TP7	4V-5V when data is received

CON4. For this, set data bits AD8 to AD11 to 1000 using DIP2. Push all three pins 11, 12 and 13 (except pin 10) of IC1 to ground. Now, to open the lock, power on the receiver circuit using S2; D8 through D11 should go low.

Next, switch on the transmitter circuit. In the receiver, D8 of IC2 will go high and transistor T3 will conduct, which, in turn, will open solenoid lock1 at CON4. You can remotely operate the solenoid lock from a distance of 30 metres or more depending on the RF modules used. **EFY**



Shashwat Mishra is pursuing BE from BIT Mesra. His interests include designing electronics circuits